

Stage-by-Stage Performance Analysis (LKA ↔ AEB Endurance)

Stage 1 – Straight Road Lane Keeping

During the straight-road section, the Lane Keeping Assist (LKA) module autonomously maintained lane tracking using camera-based lane detection and PID-based steering control. The vehicle maintained stable lane positioning with smooth steering corrections and minimal oscillation.

The PI-based longitudinal controller maintained the target speed close to **30 km/h** throughout this stage.

Observed Performance

- Stable lane tracking during straight-road driving.
 - Smooth steering response with minimal oscillation.
 - Vehicle speed maintained near the target value.
 - No unnecessary braking or instability observed.
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Stage 2 – Curved Road Navigation

In the curved-road section, the LKA controller continued operating autonomously without any external steering assistance. The vehicle successfully followed the curved path using image-based lane tracking.

Due to the single-lane edge tracking approach, the controller produced a near-steady steering correction during sustained curvature. However, the vehicle remained stable and within lane boundaries throughout the curved section.

Observed Performance

- Successful autonomous curve negotiation.
- Stable steering behaviour during sustained curvature.
- Bounded lateral deviation (tRoad) throughout the run.
- No lane departure observed.

Limitations Observed

- Steering values remained biased toward one direction during long curves.
 - Camera feed visually rotated due to continuous vehicle yaw.
 - Single-lane tracking introduced steady-state steering bias.
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Stage 3 – Dynamic Object Insertion (AEB)

During the dynamic object insertion stage, the radar-based AEB system continuously monitored:

- Relative Distance (Dist)

- Relative Velocity (V_{rel})
- Lateral Offset ($DistY$)

When the target vehicle entered the ego lane and violated the safety threshold, the AEB system triggered automatic braking.

The throttle command was forced to zero and braking was applied autonomously. A persistent AEB state machine was implemented to avoid repeated stop-go-stop behaviour.

The ego vehicle remained stopped until the obstacle fully exited the ego lane, after which autonomous cruise control resumed.

Observed Performance

- Reliable radar-based target detection.
- Successful emergency braking before collision.
- Stable stop-and-hold behaviour.
- Smooth autonomous resume after obstacle clearance.

LKA ↔ AEB Mode Transition Analysis

The endurance controller integrated both LKA and AEB modules into a single autonomous driving system.

- LKA remained continuously active during all stages.
- AEB temporarily overrode only the longitudinal control during hazard detection.
- Steering control continued operating even during braking events.
- Once AEB triggered:
 - throttle was set to zero,
 - braking was applied automatically,
 - vehicle remained stopped until safe conditions returned.
- After obstacle clearance, autonomous cruise control resumed smoothly.

This transition enabled stable lane tracking during braking and ensured safe integrated behaviour throughout the endurance scenario.

Overall System Evaluation

The integrated endurance controller successfully demonstrated:

- autonomous lane keeping,
- autonomous speed regulation,
- emergency braking,
- obstacle handling,

- and autonomous resume behaviour.

The system completed all endurance stages while maintaining stable lateral and longitudinal vehicle control.

Key Achievements

- Successful integration of LKA and AEB.
- Stable autonomous cruising near **30 km/h**.
- Reliable radar-based emergency braking.
- Proper stop-and-resume functionality.
- Continuous autonomous steering throughout the endurance run.

Future Improvements

- Two-lane lane-center tracking for improved steering symmetry.
- Advanced path-tracking algorithms for reduced steering bias.
- Improved camera stabilization and perception robustness.